

SWAMI VIVEKANANDA, India

WMO: 428750

Lat: 21.183N

Lon: 81.733E

Elev: 1042

StdP: 14.15

Time Zone: 5.50 (IND)

Period: 82-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	53.7	56.0	32.7	28.2	80.9	36.5	32.9	80.1	13.5	73.5	11.4	70.2	1.5	0	0.318

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	22.6	109.8	72.9	107.3	73.1	104.9	72.9	81.2	91.8	80.4	90.0	79.8	88.6	5.6	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.1	157.3	84.4	78.4	153.6	83.7	77.8	150.1	83.1	46.1	92.0	45.2	90.9	44.4	88.6	90.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.9	12.6	11.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual														
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	80.9	69.8	74.2	82.7	90.4	94.8	89.6	82.5	81.3	82.1	80.3	73.3	69.1	(6)
(7)		DBStd	8.70	3.59	3.81	4.67	3.91	4.63	6.05	3.33	2.60	2.20	2.93	3.34	3.15	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	13	8	0	0	0	0	0	0	0	0	0	0	5	(9)
(10)		CDD50	11267	613	678	1015	1211	1389	1189	1008	970	962	939	700	593	(10)
(11)		CDD65	5804	156	258	550	761	924	739	543	505	512	474	250	132	(11)
(12)		CDH74	72489	1197	2423	7059	11777	15409	10806	5926	5006	5396	4593	1884	1013	(12)
(13)		CDH80	38386	218	936	3993	7719	11026	6750	2176	1422	1870	1654	465	157	(13)
(14)	Wind	WSAvg	3.8	1.8	2.8	3.2	4.7	4.3	6.2	5.7	5.1	4.1	2.9	2.5	2.7	(14)
(15)	Precipitation	PrecAvg	48.7	0.4	0.5	0.7	0.5	0.8	6.9	14.5	13.1	8.6	2.2	0.2	0.3	(15)
(16)		PrecMax	64.8	4.3	2.8	1.9	4.4	4.4	24.9	27.5	20.5	17.5	6.9	2.4	2.6	(16)
(17)		PrecMin	31.3	0.0	0.0	0.1	0.0	0.0	0.9	4.3	6.4	1.6	0.1	0.0	0.0	(17)
(18)		PrecStd	9.1	0.9	0.7	0.4	0.9	1.0	4.8	5.6	3.8	3.8	2.0	0.5	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.4	93.6	102.6	108.5	114.3	112.1	98.8	91.7	92.2	92.9	87.8	85.7	(19)
(20)		MCWB	62.5	65.1	69.4	70.2	71.5	77.6	76.9	79.1	76.8	74.3	69.0	65.7	(20)	
(21)		2%	DB	84.2	91.0	100.3	106.2	111.5	107.4	94.3	89.6	90.9	91.0	86.1	83.6	(21)
(22)		MCWB	63.5	64.9	68.1	70.9	71.5	75.4	78.6	79.1	77.2	74.2	67.5	64.9	(22)	
(23)		5%	DB	82.4	88.2	98.4	104.5	108.9	104.1	91.0	88.0	89.6	89.3	84.6	81.9	(23)
(24)		MCWB	63.5	64.9	68.4	70.4	72.7	75.9	79.1	78.9	77.4	74.6	66.8	64.2	(24)	
(25)		10%	DB	80.6	85.9	96.2	102.6	106.6	100.7	88.7	86.3	88.1	87.7	83.3	80.0	(25)
(26)		MCWB	63.4	64.5	68.0	70.5	73.4	76.6	78.9	78.5	77.5	74.9	66.3	63.3	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.6	73.1	76.8	80.2	82.3	82.4	82.7	81.0	80.6	79.6	75.0	70.0	(27)
(28)		MADB	80.9	86.5	95.6	99.7	105.8	97.9	90.1	86.2	87.1	85.8	80.0	80.8	(28)	
(29)		2%	WB	68.8	70.9	73.1	77.2	79.8	81.0	81.2	80.2	79.8	78.7	72.9	68.2	(29)
(30)		MADB	78.9	82.8	92.0	97.9	100.4	95.5	88.3	85.9	86.3	85.4	79.7	78.8	(30)	
(31)		5%	WB	67.5	69.3	71.1	75.3	78.3	80.0	80.3	79.7	79.2	78.0	71.6	66.9	(31)
(32)		MADB	77.6	80.2	92.2	96.7	98.2	93.4	87.1	85.3	85.5	84.5	79.4	77.7	(32)	
(33)		10%	WB	66.2	67.7	69.7	73.6	76.8	79.2	79.5	79.1	78.7	77.3	70.0	65.4	(33)
(34)		MADB	75.8	78.7	89.9	95.1	97.4	91.3	85.9	84.4	85.0	83.6	78.5	76.2	(34)	
(35)	Mean Daily Temperature Range	MDBR	21.9	22.6	24.0	23.8	22.6	17.5	10.5	8.7	11.3	14.9	20.5	22.6	(35)	
(36)		5% DB	MADB	23.8	26.0	26.3	24.9	24.4	22.0	14.8	11.5	13.7	16.4	21.2	24.3	(36)
(37)		MCWBR	8.9	9.0	9.2	7.9	7.5	5.5	4.6	3.6	4.2	5.3	6.8	9.4	(37)	
(38)		5% WB	MADB	18.3	19.4	24.0	23.0	22.6	18.3	12.1	10.2	11.6	13.1	15.2	19.9	(38)
(39)		MCWBR	8.7	9.4	10.7	10.2	9.1	6.2	4.9	4.1	4.6	5.0	5.9	8.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.465	0.475	0.536	0.555	0.600	0.656	0.594	0.555	0.549	0.528	0.513	0.472	(40)	
(41)		taud	2.022	2.015	1.888	1.889	1.815	1.738	1.883	2.007	2.006	1.998	1.948	2.016	(41)	
(42)		Ebn at Noon	252	259	248	245	232	218	232	242	242	240	235	244	(42)	
(43)		Edh at Noon	51	54	64	64	69	74	64	56	56	54	54	49	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1436	1698	1955	2162	2170	1654	1255	1239	1479	1595	1468	1365	(44)	
(45)		RadStd	75	73	94	80	50	167	151	118	141	128	81	70	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page